

Geometric Selection of the Leptonic CP Phase from Platonic Base Angles and a Discrete δ_{CP} Lattice

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We present a discrete-geometric selection principle for the PMNS mixing parameters, emphasizing the leptonic Dirac CP phase δ_{CP} . The approach mirrors a CKM-style “profile scan + evidence weighting” methodology, but uses only Platonic base angles and an *a priori* discrete lattice for δ_{CP} , rather than continuously fitting δ_{CP} . The continuous degrees of freedom are restricted to small multiplicative symmetry-breaking corrections $\theta_{ij} = \theta_{ij}^0(1 + \varepsilon_{ij})$ around fixed Platonic angles $\theta_{23}^0 = \arcsin(1/\sqrt{2})$, $\theta_{12}^0 = \arcsin(1/\sqrt{3})$, $\theta_{13}^0 = \arcsin(1/\sqrt{45})$. Using NuFIT v5.2 (November 2022) one-dimensional $\Delta\chi^2$ profiles for normal ordering as input likelihoods, we perform a full profile scan over the discrete δ_{CP} candidates (divisors of 360° plus two explicit Platonic candidates). The global best fit selects $\delta_{CP} \approx 232^\circ$, lying within 0.24° of the Platonic value $\delta_2 = \pi + \arcsin(\sqrt{10}/4) = 232.2388^\circ$, with $\sin \delta_2 = -\sqrt{10}/4$, $\cos \delta_2 = -\sqrt{6}/4$. Posterior predictive checks and a δ -holdout test support the robustness of the δ_{CP} selection under the factorized-likelihood approximation. This yields a sharp, falsifiable target for ongoing and upcoming long-baseline neutrino programs.

Caveat: the likelihood is approximated by summing 1D $\Delta\chi^2$ profiles; correlations are neglected.

Motivation.—The Standard Model extended to include neutrino masses accommodates leptonic flavor mixing through the PMNS matrix, yet the observed mixing pattern—particularly the Dirac CP phase δ_{CP} —remains only partially determined. Global analyses (e.g., NuFIT [1]) summarize current oscillation data as likelihood surfaces over $\sin^2 \theta_{12}$, $\sin^2 \theta_{13}$, $\sin^2 \theta_{23}$, and δ_{CP} . In this work we test an organizing principle: mixing parameters are selected by discrete geometry rather than continuously tuned.

This study continues a broader program in which geometric templates constrain Standard Model flavor structures. Companion work shows (i) a parameter-free geometric Hamiltonian reproducing charged-lepton mass ratios using fixed geometric anchors $t_i/t_r = 180/\pi$ and $v/|t_i| = \sqrt{15}$ [2], and (ii) a CKM selection rule in which discrete hypotheses are declared *a priori* and tested via profile scans and evidence weights [3].

Here we apply an analogous “discrete hypothesis + profile selection” strategy to PMNS, focusing on the selection of δ_{CP} .

Geometric hypotheses.—We define the Platonic base angles:

$$\theta_{23}^0 = \arcsin\left(\frac{1}{\sqrt{2}}\right) = 45^\circ, \quad (1)$$

$$\theta_{12}^0 = \arcsin\left(\frac{1}{\sqrt{3}}\right) = 35.2644^\circ, \quad (2)$$

$$\theta_{13}^0 = \arcsin\left(\frac{1}{\sqrt{45}}\right) = 8.5731^\circ. \quad (3)$$

The physical angles are restricted to small multiplicative deformations,

$$\theta_{ij} = \theta_{ij}^0(1 + \varepsilon_{ij}), \quad (4)$$

with conservative bounded ranges for ε_{ij} (used in the numerical scan).

We consider a small set of “model classes” that determine which ε_{ij} are free parameters (continuous) and whether θ_{23} is constrained to an octant. In particular, the scan includes models where ε_{13} is fixed to zero as well as a model where ε_{13} is released (denoted $M_{\text{free}23+\text{free}13}$ in the implementation).

Discrete δ_{CP} lattice.—A central design choice is that δ_{CP} is not treated as a continuous fit parameter. Instead, δ_{CP} is selected from an *a priori* discrete lattice:

Equal-division lattice: $\delta = k(360^\circ/n)$ with $n \in \{12, 15, 18, 20, 24, 30, 36, 40, 45, 60, 72\}$, including all $k = 0, \dots, n-1$.

Explicit Platonic candidates:

$$\delta_1 = \pi + \arcsin(1/\sqrt{12}) = 196.7787^\circ, \quad (5)$$

$$\delta_2 = \pi + \arcsin(\sqrt{10}/4) = 232.2388^\circ, \quad (6)$$

which are injected into the lattice even though they are not integer-degree points.

For δ_2 , the trigonometric values are exactly

$$\sin \delta_2 = -\frac{\sqrt{10}}{4}, \quad \cos \delta_2 = -\frac{\sqrt{6}}{4}, \quad (7)$$

since $(10 + 6)/16 = 1$.

Data.—As input constraints we use NuFIT v5.2 (data available in November 2022) for normal ordering, taking the one-dimensional $\Delta\chi^2$ profiles for:

$$\sin^2 \theta_{12}, \sin^2 \theta_{13}, \sin^2 \theta_{23}, \delta_{CP}. \quad (8)$$

NuFIT publicly documents the v5.2 (Nov 2022) fit and provides χ^2 data tables (from which 1D projections can be extracted) [4].

In our implementation, each 1D profile is ingested as a CSV with columns $(x, \Delta\chi^2)$, where $x = \sin^2 \theta_{ij}$ for the angles and $x = \delta$ (degrees) for δ_{CP} .

Likelihood proxy and profile scan.—Given a candidate $(\theta_{12}, \theta_{13}, \theta_{23}, \delta)$, we compute

$$\chi_{1D}^2(\theta_{12}, \theta_{13}, \theta_{23}, \delta) = \Delta\chi_{12}^2(\sin^2 \theta_{12}) + \Delta\chi_{13}^2(\sin^2 \theta_{13}) + \Delta\chi_{23}^2(\sin^2 \theta_{23}) + \Delta\chi_\delta^2(\delta), \quad (9)$$

where each $\Delta\chi^2$ is obtained by linear interpolation on the NuFIT 1D grid, with an edge penalty outside the provided support.

This corresponds to a factorized-likelihood approximation (correlations neglected), which we state explicitly as a caveat.

For each discrete δ in the lattice and each model class, we minimize χ_{1D}^2 over the relevant ε_{ij} parameters using bounded multi-start local optimization (Powell).

We then form a profile function over δ , and define normalized evidence-like weights,

$$w(\delta) \propto \exp\left[-\frac{1}{2}(\chi_{\text{prof}}^2(\delta) - \chi_{\text{min}}^2)\right], \quad \sum_{\delta} w(\delta) = 1. \quad (10)$$

This mirrors the CKM discrete-selection strategy used in the companion CKM study [3].

We report χ^2 minima and, treating δ as a discrete index (not a continuous parameter), compute AIC/BIC with $n_{\text{obs}} = 4$ (one per 1D profile) and k equal to the number of continuous ε parameters in the model class.

Two robustness tests are included:

δ -holdout: fit only the three angle profiles (exclude $\Delta\chi_\delta^2$), then score each discrete δ using $\Delta\chi_\delta^2(\delta)$.

Posterior predictive checks (PPC): draw synthetic “toy” central values from $\propto \exp(-\Delta\chi^2/2)$ on each 1D grid, recenter the curves so the toy has $\Delta\chi^2 = 0$, and rerun the full discrete selection on the same δ lattice.

Results.—Using the above pipeline, the global best fit is obtained in the model class allowing ε_{13} to float ($M_{\text{free23+free13}}$), selecting the lattice point

$$\delta_{CP} = 232^\circ, \quad (11)$$

with an effectively vanishing χ^2 under the 1D-profile sum (numerically $\chi^2 \sim 3 \times 10^{-11}$, consistent with the minimum of the 1D curves). This selected value lies within 0.2388° of the Platonic candidate $\delta_2 = 232.2388^\circ$ in Eq. (6).

The corresponding fitted deformations are:

$$\begin{aligned} \varepsilon_{12} &= -0.049388, \\ \varepsilon_{23} &= -0.062489, \\ \varepsilon_{13} &= +0.001762, \end{aligned} \quad (12)$$

yielding angle-level predictions (converted from Eq. (4)):

$$\begin{aligned} \theta_{12} &\approx 33.523^\circ, \\ \theta_{23} &\approx 42.188^\circ, \\ \theta_{13} &\approx 8.588^\circ, \end{aligned} \quad (13)$$

or equivalently

$$\begin{aligned} \sin^2 \theta_{12} &\approx 0.3050, \\ \sin^2 \theta_{23} &\approx 0.4510, \\ \sin^2 \theta_{13} &\approx 0.02230. \end{aligned} \quad (14)$$

Discrete evidence over δ . The top-weighted δ candidates cluster tightly around 232° and include the explicit Platonic candidate δ_2 itself (as a non-integer-degree lattice point). This behavior is expected when the best-fit region is narrow relative to the full discrete lattice: normalized weights are distributed across the nearest neighbors.

Information criteria. Because $n_{\text{obs}} = 4$ is small, BIC can strongly penalize the additional continuous parameter in $M_{\text{free23+free13}}$ relative to two-parameter classes; we therefore report AIC/BIC as conservative summaries rather than as the primary selection statistic.

Robustness tests. In the δ -holdout, the δ scoring continues to favor the same neighborhood, indicating that the δ selection is not an artifact of jointly fitting angles and phase.

In PPC, repeated toy experiments re-select the same geometric model class and δ region under the factorized-likelihood procedure.

The model comparison results are summarized in Table I, and the best-fit parameters are given in Table II.

TABLE I. Model comparison for discrete δ_{CP} selection. The model $M_{\text{free23+free13}}$ achieves an effectively vanishing χ^2 , though BIC penalizes the additional parameter given the small $n_{\text{obs}} = 4$.

Model	χ^2	AIC	BIC	w_{BIC}
M_{free23}	9.6×10^{-2}	4.10	2.87	37.9%
$M_{\text{lower octant}}$	9.6×10^{-2}	4.10	2.87	37.9%
$M_{\text{free23+free13}}$	3.4×10^{-11}	6.00	4.16	19.9%
$M_{\text{maximal23}}$	7.10	9.10	8.48	2.3%
$M_{\text{upper octant}}$	5.90	9.90	8.67	2.1%

TABLE II. Best-fit PMNS parameters from the geometric selection. Platonic base angles are defined *a priori*; fitted values result from small multiplicative deformations $\theta_{ij} = \theta_{ij}^0(1 + \varepsilon_{ij})$. The CP phase δ_{CP} is selected from the discrete lattice.

Parameter	Platonic (<i>a priori</i>)	Best fit	ε
θ_{12}	$\arcsin(1/\sqrt{3}) = 35.26^\circ$	33.52°	-0.049
θ_{23}	$\arcsin(1/\sqrt{2}) = 45.00^\circ$	42.19°	-0.062
θ_{13}	$\arcsin(1/\sqrt{45}) = 8.57^\circ$	8.59°	+0.002
δ_{CP}	$\pi + \arcsin(\sqrt{10}/4) = 232.24^\circ$	232°	(discrete)

Discussion.—*Factorization approximation and robustness:* In this work we approximate the global likelihood

by a sum of one-dimensional $\Delta\chi^2$ profiles provided by NuFIT v5.2 (Nov. 2022), thereby neglecting correlations among oscillation parameters. This limitation is explicit and stems from the public availability of the likelihood information. Importantly, this approximation retains the full non-Gaussian structure of each marginal profile and therefore constitutes a strict improvement over Gaussian fallback treatments. Moreover, the aim of the present analysis is not a high-precision determination of continuous parameters, but the selection of discrete geometric hypotheses for δ_{CP} . In this context, correlations may shift the exact location of the minimum at the level of a few degrees, but are unlikely to remove or generate isolated minima in the discrete δ lattice. This expectation is supported by δ -holdout tests and posterior predictive checks, which consistently select the same δ_{CP} region across model classes. A future extension using full multidimensional likelihoods, when available, would provide a sharper test of the geometric selection rule.

Falsifiability: A key distinction from retrospective “reconstruction” of already-fixed parameters is that the leptonic CP phase remains an active target of precision measurement in long-baseline oscillation experiments. T2K continues to publish constraints on oscillation parameters including δ_{CP} [5].

Future facilities such as Hyper-Kamiokande are explicitly designed to measure leptonic CP violation with substantially improved precision, providing a direct test of any sharp δ_{CP} target [6].

In this context, the main output of the present work is naturally framed as a discrete geometric target:

$$\delta_{CP} \text{ is selected near } \delta_2 = \pi + \arcsin(\sqrt{10}/4) \approx 232.24^\circ, \quad (15)$$

rather than as a continuously tuned best-fit parameter. As experimental uncertainties shrink, the discrete profile landscape over δ will either stabilize on this candidate (supporting the selection rule) or move away (falsifying it), paralleling the philosophy articulated in the CKM discrete-selection analysis [3].

The present analysis uses NuFIT v5.2 likelihoods based on data available in November 2022. While updated global fits will refine the preferred regions, the discrete-selection framework presented here is designed to be di-

rectly reapplied as new likelihood releases become available.

Conclusion.—We have implemented a discrete-geometric selection analysis for PMNS mixing using (i) Platonic base angles, (ii) small bounded symmetry-breaking deformations, and (iii) an *a priori* discrete δ_{CP} lattice augmented by two explicit Platonic candidates. Using NuFIT v5.2 (Nov 2022) 1D $\Delta\chi^2$ profiles for normal ordering, a full profile scan selects $\delta_{CP} \approx 232^\circ$, within 0.24° of $\delta_2 = \pi + \arcsin(\sqrt{10}/4)$. Holdout and PPC diagnostics support robustness within the factorized-likelihood approximation. The resulting δ_{CP} target is sharply testable by ongoing and upcoming long-baseline neutrino experiments.

Data and code availability.—The analysis code and NuFIT v5.2 1D $\Delta\chi^2$ profile data used in this work are publicly available at https://github.com/miosync-masa/Pmns_geometric.

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